

Fig. 48.1 Renin-angiotensin-aldosterone system (RAAS)-effect site for each type of RAAS-blocking drugs. The two major classes of drugs that target the RAAS are the angiotensin-converting enzyme (ACE) inhibitors and the selective AT1 receptor blockers (ARBs). Both these drug classes target angiotensin II, but the differences in their mechanisms of action have implications for their effects on other pathways and receptors. Both ACE inhibitors and ARBs are effective antihypertensive agents that have been shown to reduce the risk of cardiovascular and kidney events. Direct inhibition of renin—the most proximal aspect of the RAAS—became clinically feasible in 2007 with the introduction of aliskiren. AT, Angiotensin; DRI, direct renin inhibitor. (From Robles NR, Careco J, Hernandez-Gallego R. Renin-angiotensin system blocking drugs. *J Cardiovasc Pharmacol Ther*. 2014;19:14–33.)

Fig_48_1

Studies	Intervention		Control		RR (95% CI) per 10 mm Hg reduction in systolic blood pressure
	Events	Participants	Events	Participants	
Major cardiovascular events	55	13,209	137,319	14,068	0.80 (0.77–0.83)
Coronary heart disease	56	4862	136,986	5301	0.83 (0.78–0.88)
Stroke	54	4635	136,682	5378	0.73 (0.68–0.77)
Heart failure	43	3284	115,411	3760	0.72 (0.67–0.78)
Renal failure	16	890	39,888	834	0.95 (0.84–1.07)
All-cause mortality	57	9775	138,298	9998	0.87 (0.84–0.91)

Fig. 48.4 Group meta-analysis of the efficacy of blood pressure lowering with antihypertensive medication for prevention of cardiovascular disease and all-cause mortality. The results are based on pooling of the findings in 123 randomized controlled trials, with an overall sample size of 613,815 persons, and have been standardized to depict the effect of a 10 mm Hg lowering of systolic blood pressure. (Reproduced with permission from Ettehad D, Emdin CA, Kiran A, et al. *Lancet*. 2016;387:957–967.)

Fig_48_4

Table 48.1 Pharmacologic Classification of Nondrugs Antihypertensive Drugs
Angiotensin-converting enzyme inhibitors Sulfonyl Carboxyl Phosphoryl Angiotensin II type 1 receptor antagonists Biphenyl tetrazoles Nonbiphenyl tetrazoles Nonheterocyclic β-Adrenergic antagonists and α ₁ - and β-adrenergic antagonists Nonselective β-adrenergic antagonists Nonselective β-adrenergic antagonists with partial agonist activity β ₁ -Selective adrenergic antagonists β ₂ -Selective adrenergic antagonists with partial agonist activity Nonselective β-adrenergic and α ₁ -adrenergic antagonists Calcium channel blockers Benzothiazepines Dihydropyridines Diphenylalkylamines Central α ₂ -adrenergic agonists Central and peripheral adrenergic-neuronal blocking agents Direct-acting vasodilators Moderately selective peripheral α ₁ -adrenergic antagonists Peripheral α ₁ -adrenergic antagonists Peripheral adrenergic-neuronal blocking agents Renin inhibitors Aldosterone receptor antagonists Tyrosine hydroxylase inhibitors Dual endothelin A and B receptor antagonists

Table_48_1

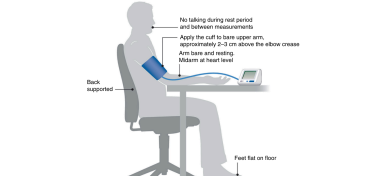
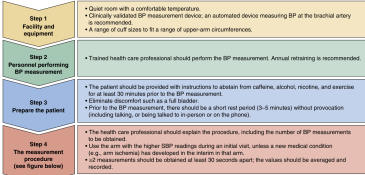


Fig. 48.2 Core elements of a standardized approach to office-based measurement of blood pressure. (Reproduced with permission from Cheung WK, Whelton PK, Martinez R, et al. *Am J Med*. 2013;126:428–435.)

Fig_48_2

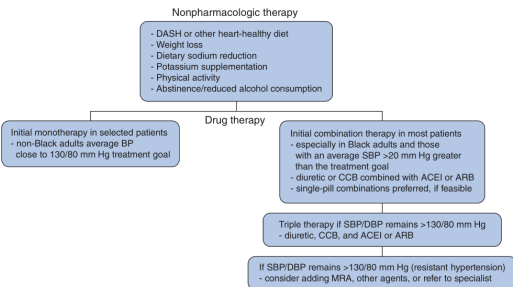


Fig. 48.5 Antihypertensive treatment in adults with uncomplicated hypertension. ACEI, Angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BP, blood pressure; CCB, calcium channel blocker; DASH, dietary approaches to stop hypertension; DBP, diastolic BP; MRA, mineralocorticoid receptor antagonists; SBP, systolic BP.

Fig_48_5

Table 48.2 Antihypertensive Mechanisms of Action of Angiotensin-Converting Enzyme Inhibitors

Lower peripheral vascular resistance.
Inhibit the breakdown of vasodilatory bradykinins.
Enhance vasodilatory prostaglandin synthesis.
Improve nitric oxide-mediated endothelial function.
Reverse vascular hypertrophy.
Decrease aldosterone secretion.
Induce natriuresis.
Augment renal blood flow.
Blunt sympathetic nervous system activity and pressor responses.
Inhibit norepinephrine and arginine vasopressin release.
Inhibit baroreceptor reflexes.
Reduce endothelin-1 levels.
Inhibit thirst.
Inhibit oxidation of cholesterol.
Inhibit collagen deposition in target organs.

Table_48_2

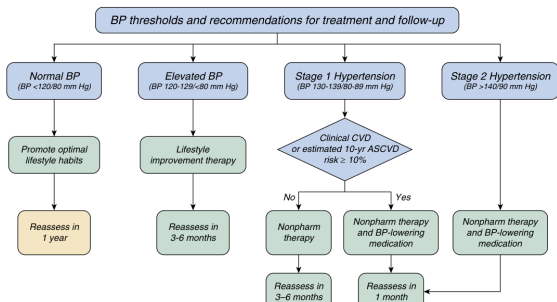


Fig. 48.3 American College of Cardiology/American Heart Association recommendations for treatment and frequency of follow-up in adults by category of blood pressure. (Reprinted with permission from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APHA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: A report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:e13–e115. © 2017 American Heart Association, Inc.)

Fig_48_3

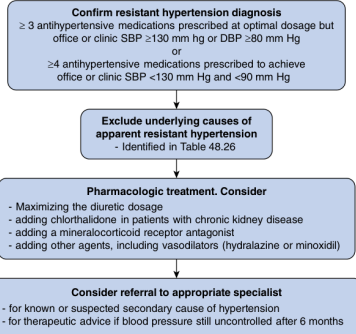


Fig. 48.6 Approach to management of apparent resistant hypertension.

Fig_48_6

Table 48.3 Pharmacodynamic Properties of Angiotensin-Converting Enzyme Inhibitors						
Generic Name (Trade Name)	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Alacepril (Cetapril)	12.5	12.5–100	100	qd	3	24
Benazepril (Lotensin)	10	20–40	40	qd	2–6	24
Captopril (Capoten)	25	12.5–50	150	tid	1–2	3–8
Cilazapril (Dynamil)	2.5	2.5–10	10	qd, bid	6	8–12
Enalapril (Vasotec)	5	10–40	40	qd, bid	3–4	12–24
Fosinopril (Monopril)	5	5–40	40	qd, bid	2–7	24
Imidapril (Tanatril)	5	10–40	40	qd	5–6	24
Lisinopril (Zestril, Priviliv)	10	20–40	40	qd	6	24
Perindopril (Aceon)	4	4–8	8	qd	3–7	24
Quinapril (Accupril)	10	20–80	80	qd	2	24
Ramipril (Altace)	2.5	2.5–20	40	qd, bid	2	24
Spirapril (Renomax)	6	6	6	qd	3–6	24
Trandolapril (Mavik)	1	2–4	8	qd	2–12	24
Zofenopril (Bifril)	15	30–60	60	qd	—	—

Table_48_3

Table 48.4 Pharmacokinetic Properties of Angiotensin-Converting Enzyme Inhibitors								
Drug	Absorption (%)	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Alacepril	—	70	—	1	1.9	L	U (70%)	Captopril
Benzazepril	35	>37	No	2–6	22	L, K	F, U	Benazeprilat
Captopril	60–75	75	Yes	1	2	L	U	Inactive
Cilazapril	—	57–76	No	1–2	30–50	L	U (52%)	Cilazaprilat
Enalapril	55–75	73	—	3–4	11–35	L	F, U	Enalaprilat
Fosinopril	36	36	—	1	12	L, K, I	F, U	Fosinoprilat
Imidapril	—	40	—	3–10	10–19	L	U	Imidaprilat
Lisinopril	25	6–60	—	1	12	K	U	Enalaprilat
Quinapril	60	50	Yes	1	25	L	U (50%)	Quinaprilat
Perindopril	—	75	Yes	1.5	3–10	L	F, U (75%)	Perindoprilat
Ramipril	50–60	60	—	1–2	13–17	L	F, U	Ramiprilat
Spirapril	—	50	Yes	1	33–41	L	F (60%), U (40%)	Spiraprilat
Trandolapril	70	10	No	2–12	16–24	L	F (86%), U	Trandolaprilat
Zofenopril	>80	96	Yes	5	5	K	F (26%), U (69%)	Zofenoprilat

F, Feces; I, intestine; K, kidney; L, liver; U, urine.

Table_48_4

Table 48.5 Renal Protective Mechanisms of Angiotensin-Converting Enzyme Inhibitors

Decrease renal blood flow.
Decrease arterial pressure.
Decrease filtration fraction.
Decrease renal vascular resistance.
Decrease aldosterone production.
Decrease proteinuria.
Inhibit tubule sodium resorption.
Restore pressure-natriuresis relationship.
Improve altered lipid profile.
Reduce scarring and fibrosis.
Attenuate oxidative stress and reduce free radicals.

Table_48_5

Table 48.8 Dose Modifications of Antihypertensive Drugs Required for Reduced Kidney Function				
Estimated Glomerular Filtration Rate (Creatinine Clearance; mL/min/1.73 m ²)				
Drug	>50	10–50	<10	Dialysis ^a
Angiotensin-Converting Enzyme Inhibitors				
Alacepril	No change	50%	25%	Hepatic
Captopril	No change	50%	25%	± 50%
Enalapril	No change	50%	25%	± 50%
Fosinopril	No change	No change	15%	—
Imidapril	No change	No change	25%	± 50%
Lisinopril	No change	50%	25%	—
Perindopril	No change	50%	25%	—
Quinapril	No change	50%	25%	—
Trandolapril	No change	50%	25%	—
Zofenopril	No change	50%	25%	—
Angiotensin Receptor Blockers				
Azilsartan	No change	No change	No change	Hepatic
Candesartan	No change	No change	50%	Hepatic
Eprosartan	No change	No change	No change	Hepatic
Irbesartan	No change	No change	No change	Hepatic
Losartan	No change	No change	No change	Hepatic
Olmesartan	No change	No change	No change	Hepatic
Telmisartan	No change	No change	No change	Hepatic
Valsartan	No change	No change	No change	Hepatic
β-Blockers				
Acetaminophen	No change	50%	50%	± 50%
Albuterol	No change	50%	50%	± 50%
Bisoprolol	No change	50–75%	25%	± 50%
Carvedilol	No change	50%	25%	Hepatic
Clonidine	No change	No change	No change	—
Diltiazem	No change	50%	25%	—
Enalapril	No change	No change	No change	—
Fosinopril	No change	50%	25%	Hepatic
Hydrochlorothiazide	No change	No change	No change	—
Metoprolol	No change	No change	50%	Hepatic
Nifedipine	No change	No change	50%	Hepatic
Propranolol	No change	No change	50%	Hepatic
Timolol	No change	No change	50%	Hepatic
Calcium Channel Blockers				
Amlodipine	No change	No change	No change	Hepatic
Diltiazem	No change	No change	No change	Hepatic
Felodipine	No change	No change	No change	Hepatic
Isradipine	No change	No change	No change	Hepatic
Nifedipine	No change	No change	No change	Hepatic
Verapamil	No change	No change	No change	Hepatic
Diuretics				
Acetaminophen	No change	No change	No change	Hepatic
Albuterol	No change	No change	No change	Hepatic
Bisoprolol	No change	No change	No change	Hepatic
Carvedilol	No change	No change	No change	Hepatic
Clonidine	No change	No change	No change	Hepatic
Diltiazem	No change	No change	No change	Hepatic
Enalapril	No change	No change	No change	Hepatic
Fosinopril	No change	No change	No change	Hepatic
Hydrochlorothiazide	No change	No change	No change	Hepatic
Metoprolol	No change	No change	No change	Hepatic
Nifedipine	No change	No change	No change	Hepatic
Propranolol	No change	No change	No change	Hepatic
Timolol	No change	No change	No change	Hepatic
Central α ₁ -Adrenergic and Serotonin Receptor Agonists				
Clonidine	No change	50%	25%	Hepatic
Enalapril	No change	50%	25%	± 50%
Losartan	No change	50%	25%	—
Metoprolol	No change	50%	25%	—

Continued on following page

Table_48_6_part1

Table 48.6 Dose Modifications of Antihypertensive Drugs Required for Reduced Kidney Function (Cont'd)

Estimated Glomerular Filtration Rate (Creatinine Clearance; mL/min/1.73 m ²)				
Drug	>50	10–50	<10	Dialysis ^a
Direct-Acting Vasodilators				
Hydralazine	No change	No change	75% ^b	Negligible
Minoxidil	No change	50%	50%	H and P: 50%
Peripheral Adrenergic-Neuronal Blocking Agents				
Guanethidine	No change	No change	—	—
Guanadrel	No change	50%	25% (avoid)	—
Renin Inhibitor				
Aliskiren	No change	No change	Not studied	Not studied
Tyrosine Hydroxylase Inhibitor				
Metoprolol	No change	50%	25%	—
Selective Aldosterone Receptor Antagonist				
Eplerenone	Dosage adjustment in renal failure unknown	—	—	—
	Caution regarding hyperkalemia	—	—	—

^aReplacement dose at end of dialysis as percentage of dose prescribed for patient with glomerular filtration rate <10 mL/min.
^bSlow acetylators.
Percentage of usual dose given.
K, Hemodialysis; P, peritoneal dialysis; —, not applicable.

Table_48_6_part2

Table 48.7 Pharmacodynamic Properties of Angiotensin Receptor Blockers

Generic Name (Trade Name)	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Azilsartan (Edarbi)	40	40–80	80	qd	1.5–3 peak response	≥24
Candesartan (Atacand)	16	8–32	32	qd	6–8	24
Eprosartan (Teveten)	200	400–800	800	qd, bid	4	24
Irbesartan (Avapro)	150	150–300	300	qd	4–6, 14	24
Losartan (Cozaar)	25	50–100	100	qd/bid	6	12–24
Olmesartan (Benicar)	20	20–40	40	qd	1–2	24
Telmisartan (Micardis)	40	40–80	80	qd	3	24
Valsartan (Diovan)	80	80–160	320	qd	4–6	24

bid, Two times a day; qd, once a day.

Table_48_7

Table 48.8 Pharmacokinetic Properties of Angiotensin Receptor Blockers

Drug	Absorption (%)	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Candesartan	—	15	No	2–4	9	I, L, K	F (67%), U (33%)	None
Eprosartan	>80	13	Yes	4	6	L	F (70%), U (7%)	Inactive
Irbesartan	>80	60–80	No	1.5–2	10–14	L, K	F (65%), U (20%)	Inactive
Losartan	>80	25	No	1	4–9	L, K	F (65%), U (40%)	Active
Olmesartan	—	26	No	1	13	I	F (50%), U (50%)	Active
Telmisartan	—	42	Yes	0.5–1	24	L	F	Inactive
Valsartan	>80	25	Yes	2–4	6–9	L, K	F (83%), U (13%)	Inactive

F, Feces; I, intestine; K, kidney; L, liver; U, urine.

Table_48_8

Table 48.9 Pharmacologic Properties of β-Adrenergic Antagonists

Generic Name (Trade Name)	β ₁ Selectivity	Partial Agonist Activity	Membrane-Stabilizing Activity	α-Adrenergic Antagonist Activity
Acebutolol (Sectral)	+	+	+	—
Atenolol (Tenormin)	+	—	+	—
Betaxolol (Kerlone)	+	—	+	—
Bisoprolol (Zebeta)	+	—	—	—
Carvedilol (Coreg)	—	—	+	+
Celiprolol (none in the United States)	+	—	—	+
Labetalol (Trandate)	—	+	—	+
Metoprolol (Lopressor)	+	—	—	—
Nadolol (Corgard)	—	—	—	—
Penbutolol (Levato)	—	+	+	—
Pindolol (Visken)	—	+	+	—
Propranolol (Inderal)	—	—	+	—
Nebivolol (Bystolic)	+	—	—	—

Table_48_9

Table 48.10 Pharmacokinetic properties of β-adrenergic antagonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion
Acebutolol	90	No	2–3	3–8	L	U
Atenolol	40–60	Yes	—	14–16	—	U, F
Betaxolol	78–90	No	2–6	12–22	L	U
Bisoprolol	90	—	2.3	9.6	L	U
Carvedilol	25–30	Yes	1.5	6–10	L	U, F
Metoprolol	50	—	1.5–2	3–7	L	U
Nadolol	20–40	No	2–4	20–24	—	U, F
Penbutolol	100	No	—	17–24	L	U
Pindolol	95	No	2	3–11	L	U (40%)
Propranolol	16–60	Yes	—	3–4	L	—
Timolol	50–90	—	—	2–4	L	U (20%)

F, Feces; L, liver; U, urine.

Table_48_10

Table 48.11 Pharmacodynamic Properties of β-Adrenergic Antagonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Nadolol	40	40–80	320	qd	—	—
Propranolol	40	80–320	640	bid	—	—
Timolol	10	20–40	60	bid	—	—
Penbutolol	10	20–40	80	qd, bid	2	20–24
Pindolol	5	10–40	60	qd, bid	—	24
Atenolol	25	50–100	200	qd	3	24
Metoprolol	12.5–50	100–200	400	qd, bid	1	3–6
Betaxolol	10	10–40	40	qd	3	23–25
Bisoprolol	5	2.5–20	20	qd	2–4	24
Acebutolol	400	400–800	1200	qd	3	24

bid, Two times a day; qd, once a day.

Table_48_11

Table 48.12 Pharmacokinetic Properties of β-Adrenergic Antagonists With Vasodilatory Properties							
Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Labetalol	25–40	Yes	1–2	5–8	L	U (50%–60%)	—
Carvedilol	25–35	No	1–1.5	6–8	L	F	—
Calciprolol	30–70	Yes	—	5–6	—	F, U	—
Nebivolol	12–96	No	2.4–3.1	8–27	L	—	—
F, Feces; L, liver; U, urine.							

Table_48_12

Table 48.13 Pharmacodynamic Properties of β-Adrenergic Antagonists with Vasodilatory Properties						
Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Labetalol	100	200–800	1200–2400	bid	3	8–12
Carvedilol	6.25	12.5–25	50	bid	4–7	24
Calciprolol	200	200–400	400	qd	—	—
Nebivolol	5	5	40	qd	6	24
<i>bid</i> , Two times a day; <i>qd</i> , once a day.						

Table_48_13

Table 48.14 Pharmacokinetic Properties of Calcium Channel Blockers									
Drug	Oral Absorption (%)	First-Pass Effect	Bioavailability (%)	Peak Blood Level	Elimination Half-Life (h)	Metabolism and Excretion	Protein Binding (%)	Active Metabolites	
Amlodipine	>90	M	88	6–12h	30–50	L/U	>95	Yes	
Diltiazem	98	50%	40	2–3h	4–6	L, F, U	77–93	Yes	
Diltiazem SR	>80	50%	35	6–11h	5–7	L, F, U	77–93	Yes	
Diltiazem CD	95	E	35	12h	6–8	L, F, U	77–93	Yes	
Diltiazem XR	95	E	41	4–6h	5–10	L, F, U	95	Yes	
Diltiazem ER	93	E	40–60	4–6h	10	L, F, U	95	Yes	
Felodipine	>90	E	13–18	2.5–5h	11–16	L/U	>95	No	
Isradipine	>80	E	15–25	2–3h	8	L, F, U	>95	No	
Isradipine CR	>90	E	15–25	7–18h	—	L, F, U	>95	No	
Lacidipine	>90	E	—	—	12–19	L, F, U	>95	No	
Nicardipine	>90	E	—	1.5–3	—	L, F, U	>99	No	
Manidipine	>90	E	—	2–3.5h	5–8	L, F, U	99	No	
Nicardipine	>90	E	35	0.5–2h	8.6	L, F, U	>95	No	
Nicardipine SR	>90	E	35	1–4h	—	L, F, U	>95	No	
Nifedipine	>90	20%–30%	60	<30min	2	L, U	98	Yes	
Nifedipine GITS	>90	25%–35%	86	6h	—	L, U	98	Yes	
Nifedipine ER	>90	25%–35%	86	2.5–5h	7	L, U	98	Yes	
Nisoldipine	>85	E	4–8	6–12h	10–22	L, F, U	99	No	
Verapamil	>90	70%–80%	20–35	1–2h	2.8–7.4	L, F, U	85–95	Yes	
Verapamil SR	>90	70%–80%	20–35	5–6h	4–12	L, F, U	85–95	Yes	
Verapamil SR pellet	>90	70%–80%	20–35	7–9h	12	L, F, U	85–95	Yes	
CODAS Verapamil	>90	70%–80%	20–35	11h	—	L, F, U	85–95	Yes	

CD, Controlled-diffusion; CODAS, chronotherapeutic oral drug absorption system; CR, controlled release; E, extensive; ER, extensive release; F, feces; GITS, gastrointestinal therapeutic system; L, liver; M, minimal; SR, sustained release; U, urine; XR, extended release.

Table_48_14

Table 48.15 Pharmacodynamic Properties of Calcium Channel Blockers					
Generic Name (Trade Name)	First Dose (mg)	Usual Daily Dosage (mg)	Maximum Daily Dose (mg)	Peak Response (h)	Duration of Response (h)
Diltiazem (Cardizem)	60	60–120 tid, qid	480	2.5–4	8
Diltiazem SR (Cardizem SR)	180	120–240 bid	480	6	12
Diltiazem CD (Cardizem CD)	180	240–360 qd	480	—	24
Diltiazem XR (Dilacor XR)	180	180–480 qd	480	3–6	24
Diltiazem ER (Tiazac)	180	180–480 qd	480	4–6	24
Amlodipine (Norvasc)	5	5–10 qd	10	30–50	24
Felodipine (Prestil ER)	2.5	2.5 qd	10	2–5	24
Isradipine (DynaCirc)	2.5	2.5–5 bid	20	2–3	12
Isradipine CR (DynaCirc CR)	5	5–20 qd	20	2	7–18
Lacidipine	4	4–6 qd	6	1–2	12–24
Lercandipine	10	10–20 qd	40	24	—
Manidipine	10	10–20 qd	40	2–3.5	24
Nicardipine (Cardene)	20	20–40 tid	120	0.5–2	8
Nicardipine SR (Cardene SR)	30	30–60 bid	120	1.4	12
Nifedipine (Procardia, Adalat)	10	10–30 tid, qid	120	0.1	4–6
Nifedipine GITS (Procardia XL)	30	30–90 qd	120	4–6	24
Nifedipine ER (Adalat CC)	30	30–90 qd	120	2–6	24
Nisoldipine (Sular)	20	20–40 qd	60	—	24
Verapamil (Calan, Isoptin)	80	80–120 tid	480	6–8	8
Verapamil SR (Calan SR, Isoptin SR)	120	120–240 bid	480	—	12–24
Verapamil SR Pellet (Verelan)	120	240–480 qd	480	—	24
Verapamil COER-24 (Covera-HS)	180	180–480 qhs	480	>4–5	24
bid, Two times a day; CD, Controlled-diffusion; COER, controlled-onset extended-release; CR, controlled release; ER, extended release; GITS, gastrointestinal therapeutic system; qd, once a day; qhs, every bedtime; qid, four times a day; SR, sustained release; tid, three times a day; XR, extended release.					

Table_48_15

Table 48.16 Hemodynamic Effects of Calcium Channel Blockers										
Class	Arterio-lar Dilation	Coro-nary Dilation	Cardiac After-load	Cardiac Contrac-tility	Myocar-dial O ₂ Demand	Cardiac Output	AV Con-duction	SA Auto-maticity	Heart Rate: Short Term/Long Term	Activation of Baroreceptor Reflexes
Dihydropyridines	↑↑↑	↑↑↑	↓↓	↔	↓	↑ or ↔	↔	↔	↓/↑	↑ or ↔
Diltiazem	↑↑	↑↑↑	↓	↓	↓	↔	↓	↓	↓/↑ or ↔	↔
Verapamil	↑↑	↑↑	↓	↓	↓	↔	↓	↓	↓/↑ or ↔	↔
AV, Atrioventricular; SA, sinoatrial.										

Table_48_16

Table 48.17 Drug–Drug Interactions With Calcium Channel Blockers		
Calcium Channel Blocker	Interacting Drug	Result
Verapamil	Digoxin	Digoxin level ↑ by 50%–90%
Diltiazem	Digoxin	Digoxin level ↑ by 40%
Verapamil	β-Blockers	AV nodal blockade, hypotension, bradycardia, asystole
Verapamil, diltiazem, dihydropyridines	Cyclosporine-tacrolimus and sirolimus	Increase immunosuppressive drug levels due to decreased metabolism
Verapamil, diltiazem	Cimetidine	Verapamil and diltiazem levels ↑ by decreased metabolism
Verapamil	Rifampin-phenytoin	Verapamil level ↓ by enzyme induction
Dihydropyridines	Amiodarone	Exacerbation of sick sinus syndrome and AV nodal blockade
Dihydropyridines	α-Blockers	Excessive hypotension
Dihydropyridines	Propranolol	Increases propranolol level
Dihydropyridines	Cimetidine	Increased area under the curve and plasma level of calcium channel blocker
Nicardipine	Cyclosporine	Cyclosporine level ↑ by 40%–50%
Amlodipine	Cyclosporine	Cyclosporine level ↑ by 10%
Felodipine	Flavonoids	Bioavailability ↑ by 50%
Diltiazem	Methylprednisolone	Methylprednisolone 1.25-fold
Nifedipine	Diltiazem	Nifedipine level ↑ 100%–200%
AV, Atrioventricular.		

Table_48_17

Table 48.18 Receptor Binding of Centrally Acting Antihypertensives

Drug	Receptor
Clonidine	α ₂ , I ₁
α-Methyldopa	α ₂
Guanfacine	α ₂
Rilmenidine	I ₁ > α ₂
Moxonidine	I ₁ > α ₂

I₁, Imidazole receptor; α₂, α₂-adrenergic receptor.

Table_48_18

Table 48.19 Pharmacokinetic Properties of Central Adrenergic Agonists							
Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Clonidine (Catapres)	50	—	—	6–23	L	F (30%–50%) U (24%)	Methyldopa-o-sulfite
Guanfacine (Tonex)	80	—	1–4	17	L	U (40%–75%)	—
α-Methyldopa (Aldomet)	65–96	—	1.5–5	6–23	L	F (22%) U (65%)	—
Rilmenidine (Hyperium)	80–90	No	2	2–3	L	U (90%)	—
Moxonidine (Physiotens)	100	No	0.5–3	2	L	U (90%)	—
F, Feces; L, liver; U, urine.							

Table_48_19

Table 48.20 Pharmacodynamic Properties of Central Adrenergic Agonists						
Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Clonidine	0.1	0.3–0.9	2.4	bid, tid	2–4	6–10
α-Methyldopa	250	250–500	3000	bid, tid, qid	6–9	24–48
Guanfacine	1	1–3	3	qd	6	24
Moxonidine	0.1–0.2	0.2–0.3	0.6	bid	1.5–4	48–72
Rilmenidine	1	1–2	2	qd, bid	1–2	10–12
<i>bid</i> , Two times a day; <i>qd</i> , once a day; <i>qid</i> , four times a day.						

Table_48_20

Table 48.21 Pharmacokinetic Properties of Direct-Acting Vasodilators

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Hydralazine (Apresoline)	20–50	No	1–2	1.5–8	L	U (3%–14%) F (3%–12%)	—
Minoxidil (Loniten)	90–100	—	1	4.2	L	U (90%) F (3%)	Glucuronide

F, Feces; L, liver; U, urine.

Table 48.22 Pharmacodynamic Properties of Direct-Acting Vasodilators

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Hydralazine	10	200–400	400	bid, qid	1	3–8
Minoxidil	2.5	10–20	40	qid, qid	4–8	10–12

bid, Two times a day; qd, once a day; qid, four times a day.

Table 48.23 Pharmacokinetic Properties of Peripheral α₁-Adrenergic Antagonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Doxazosin (Cardura)	62–69	No	2–5	9–22	L	F (63%–65%) U (1%–9%)	—
Prazosin (Minipress)	—	No	1–3	2–4	L	F	—
Terazosin (Hytrin)	90	Yes	1	12	L	F (45%–60%) U (10%)	—

F, Feces; L, liver; U, urine.

Table_48_21

Table_48_22

Table_48_23

Table 48.24 Pharmacodynamic Properties of Peripheral α₁-Adrenergic Antagonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Doxazosin	1	8	16	qd, qid	4–8	24
Prazosin	1	3–20	20	bid, qid	0.5–1.5	10
Terazosin	1	5	20	qd, bid	3	24

bid, Two times a day; qd, once a day; qid, four times a day.

Table 48.25 Categories of Blood Pressure in Adults

BP category	Systolic BP, mm Hg	Diastolic BP, mm Hg
Normal	<120	and <80
Elevated	120–129	and <80
Stage 1 Hypertension	130–139	or 80–89
Stage 2 Hypertension	≥140	or ≥90

Individuals with SBP and DBP in 2 categories should be designated to the higher BP category.
BP, Blood pressure (based on an average of ≥2 readings obtained on ≥2 occasions); DBP, diastolic blood pressure; SBP, systolic blood pressure.
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Table 48.26 List of the Most Common Causes of Apparent Resistant Hypertension

Nonadherence to the treatment regimen
Pseudo resistance
• White coat hypertension
• Improper measurement of blood pressure, including:
 • Use of inappropriately small cuff
Pseudo hypertension (in older adults)
Lifestyle-related causes
• Excess sodium consumption, and/or insufficient potassium intake
• Poor quality diet, overweight and obesity
• Excess alcohol consumption
• Stress and other anxiety-producing stressors
• Inactivity
Drug-related causes
• Inadequate treatment
 • Inappropriate selection of antihypertensive drugs
 • Insufficient antihypertensive drug dosage
 • Inadequate attention to lifestyle counseling
• Treatment with agents that raise blood pressure
 • Sympathomimetics
 • Nasal decongestants
 • Appetite suppressants
 • Cocaine
 • Nonsteroidal antiinflammatory agents
 • Antidepressants
 • Oral contraceptives
 • Adrenal steroids
 • Cyclosporine and tacrolimus
 • Erythropoietin and other erythropoiesis-stimulating agents
• Excess caffeine consumption
 • High intake of licorice or licorice products
Secondary causes of hypertension, including
• Endocrine disorders, especially hyperaldosteronism
• Kidney disease
• Renal artery stenosis
• Sleep apnea

Table_48_24

Table_48_25

Table_48_26

Table 48.27 Corresponding Average Values of Office/Clinic, Home, and Ambulatory Monitoring Systolic and Diastolic Blood Pressure Measurements (in mm Hg). Ambulatory Blood Pressure Measurement Values are Displayed for 24-Hour, Daytime, and Nighttime Readings

Ambulatory				
Clinic/Office	Home	24-hour	Daytime	Nighttime
120/80	120/80	115/75	120/80	100/65
130/80	130/80	125/85	130/80	110/65
140/90	135/85	130/80	135/85	135/85
160/100	145/90	145/90	145/90	140/85

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Table 48.28 Hypertensive Emergencies

Hypertensive encephalopathy
Acute aortic dissection
Central nervous system bleeding
Intracranial hemorrhage
Thrombotic cerebrovascular accident
Subarachnoid hemorrhage
Acute left ventricular failure refractory to conventional medical therapy
Myocardial ischemia or infarction associated with persistent chest pain
Accelerated or malignant hypertension^a
Toxemia of pregnancy—eclampsia^a
Renal failure or acute kidney injury
Hypertension associated with hyperadrenergic states
Pheochromocytoma
Interaction between monoamine oxidase inhibitors and tyramine-containing foods
After severe body burns
Neurogenic hypertension
Hypertension in the surgical patient^a
Associated with postoperative bleeding
After open heart or vascular surgery
Preceding emergency surgery
After kidney transplantation
Hypertension in a diabetic patient with retinal hemorrhage

^aConsidered by some authors to be a hypertensive urgency.

Table 48.29 Parenteral Drugs Used in the Treatment of Hypertensive Emergencies

Drug	Dosage	Onset of Action	Peak Effect	Duration of Action
Clevidipine	1–2 mg/h (16 mg/h maximum)	2–4 min	5–6 min	15 min
Enalaprilat	0.625–5.0-mg bolus over 5 min q6h	5–15 min	6h	6h
Esmolol	250–500 µg/kg/min × 1 (loading dose), then 50–100 µg/kg/min × 4 (maintenance); maintenance dosage may be increased to maximum of 300 µg/kg/min	1–2 min	5 min	0–30 min
Fenoldopam	0.01–1.6 µg/kg/min constant infusion	5–15 min	30 min	5–10 min
Hydralazine	10–20 mg as slow intravenous injection or 0.05–0.15 mg/min continuous infusion	1–5 min	10–80 min	3–6 h
Labetalol	20 mg bolus then 0.5 to 2 mg/minute infusion	5 min	10 min	3–6 h
α-Methyldopa	250–500 mg bolus every 6h (2 g maximum)	2–3h	3–5h	6–12h
Nicardipine	5–15 mg/h	5–10 min	45 min	50h
Nitroglycerine	5–100 µg/min infusion	1–2 min	2–5 min	3–5 min
Nitroprusside	0.25–10 µg/kg/min infusion	Immediate	1–2 min	2–5 min
Phentolamine	0.5–1.0 mg/min infusion or 2.5–5.0 mg bolus q5–15 minutes	Immediate	3–5 min	10–15 min
Trimethaphan	0.5–10 mg/min infusion bolus over 2 min (300 mg maximum)	Immediate	1–2 min	5–10 min

Table_48_27

Table_48_28

Table_48_29